

Solution Requirements

Date	11 July 2026
Team ID	xxxxxx
Project Name	StudyAI – Smart AI Study Companion & Quiz Generator
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system shall allow users to register, log in, log out, and securely authenticate using JWT-based authentication with encrypted passwords.	High
2	Authorization Levels	The system shall provide role-based access control. Students can manage their own study materials, quizzes, schedules, and analytics, while administrators can manage system data.	High
3	External Interfaces	The system shall integrate with the Groq API (Llama 3.3-70B Versatile) for AI-generated content and	High

		Firebase Firestore for cloud data storage and authentication.	
4	Transaction Processing	The system shall securely process study material uploads, AI requests, quiz submissions, study plan generation, and analytics updates in real time.	High
5	Reporting	The system shall generate dashboards showing quiz scores, learning progress, weak topics, study statistics, and allow exporting study plans and summaries.	Medium
6	Business Rules	The system shall accept only supported file formats (PDF, DOCX, TXT, and text input). AI features shall process only authenticated users' uploaded study materials.	High
7	Compliance to Laws or Regulations	The system shall use secure authentication, HTTPS communication, and follow standard data privacy and security practices for protecting user information.	Medium
8	Other	The system shall provide a responsive user interface, support desktop and mobile devices, display loading indicators, and provide meaningful error messages.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system shall load pages quickly, process AI requests efficiently, and minimize response time for uploads and content generation.	Page loads in less than 2 seconds and AI responses are generated within 10–15 seconds under normal network conditions.
2	Scalability	The system shall support an increasing number of users, study materials, and AI requests without significant performance degradation.	Supports at least 1,000 concurrent users and scalable cloud deployment.
3	Security & Data Privacy	The system shall protect user data through secure authentication, HTTPS communication, JWT-based authorization, and encrypted credential storage.	100% HTTPS communication, JWT authentication, and encrypted password storage.
4	Reliability & Availability	The system shall provide stable operation with automatic error handling and cloud-based availability.	99.9% application uptime with automatic recovery from temporary failures.
5	Usability & Accessibility	The system shall provide a responsive, user-friendly interface that works across desktop, tablet, and mobile devices.	Fully responsive design and compliance with WCAG 2.1 Level AA accessibility guidelines.
6	Other (Maintainability & Portability)	The system shall have a modular architecture that supports easy maintenance, future enhancements, and deployment across different operating	Code organized into reusable modules with support for Windows, Linux, macOS, and cloud deployment platforms.

		systems and cloud platforms.	
--	--	---------------------------------	--